METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET E-202

E-202 · Feed/effluent interchanger (cold)

DOCUMENT NO. : MEOH-V1-ME-DS-004

DOCUMENT TITLE : Datasheet E-202

PROJECT : Methanol Converter Ver 1 · OTS

STATUS : Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

E-202 · Feed/effluent interchanger (cold)

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-002.

General		
Service	Feed/effluent interchanger (cold)	MEOH-PID-002
Type	DEU: U-tubes, breech-lock channel; horizontal	TEMA (assumed)
Shells	1 (single)	
Performance (model, design point)		
Duty	36.5 MW	P.q202
Hot side in → out	206.4 → 155.9 °C	
Cold side in → out	48.6 → 96.8 °C	
LMTD / F	108.5 K / 0.965	1 shell, 2 tube passes
Model UA	336 kW/K	q202 / LMTD
Service U (with fouling)	400 W/m²K	assumed, typical
Installed area	930 m²	model UA + 10 % margin (assumed)
Design		
Tube side (effluent)	100 bar g / 280 °C	Zone B
Shell side (converter feed)	100 bar g / 170 °C	Zone B
Code	ASME VIII Div. 1 (assumed); TEMA R for exchangers	
Corrosion allowance	3 mm (CS / Cr-Mo parts), 0 (tubes)	assumed
Construction		
Tubes	2548 × OD 19.05 × 2.11 mm × 6.1 m	triangular pitch 23.8 mm
Shell inside diameter	1350 mm	tube count by TEMA bundle factors
Materials	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel (effluent); shell SA-516 Gr 70 (feed side 170 °C); tubes 304L	EQL §3
Thickness (pressure side)	51.4 + 3 → 56 mm (ID 1350)	UG-27(c)(1)
Notes		
1	Condensation of crude methanol starts on the tube side: drainable, sloped (assumed).	
2	Its cold outlet is the quench gas (TI-203).	MEOH-PID-002